PHAM THI THUY

Curriculum Vitae

Jan. 10, 1997 phamthuy10149@gmail.com github.com/thithuypham
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Academic Interests

Image Processing, Computer Vision, Machine Learning, Deep Learning, Model-based Deep Learning.

Experience

Mar 2020 - Aug 2021 — **Data Scientist** at SEEDCOM, Ho Chi Minh City, Vietnam.
— NLP: model to categorize 200k+ food/drink names of FnB market at 88 percent accuracy.
— Computer Vision: object detection model deployed on camera system to alert for messy and dirty tables at 100+ coffee stores, hence improve customer dine-in experience.
— Build report/dashboard to monitor key metrics specified by BoD. Propose key metrics specified from analytics.
— Experienced in building demand forecasting model.

Aug 2019 - Mar 2020 — **Research Assistant** at Artificial Intelligence Solution in Industrial Applications Lab (AISIA), Ho Chi Minh City, Vietnam.
Research Engineer at DeepKafe team, Ho Chi Minh City, Vietnam.

Mar 2019 - Jan 2020 — **Teaching Assistant** at Faculty of Mathematics and Computer Science, University of Science - Vietnam National University Ho Chi Minh City, Vietnam.
Course: Discrete Mathematics, Basic Programming.

Aug 2018 - Dec 2018 — **Software Engineer Intern** at R&D department of Kyanon Digital, Ho Chi Minh City, Vietnam.

Education

Sep 2021–Feb 2024 — **M.Sc. in Multimedia Engineering**, Dongguk University, Seoul, Korea. GPA - 4.5/4.5
Thesis title: *Physics-Driven Prior Learning-Based Deep Unrolling for Underwater Image Enhancement*.
Supervisor: Prof. Chul Lee.

2015–2019 — **B.Sc in Mathematics and Computer Science**, University of Science - Vietnam National University Ho Chi Minh City, Vietnam. GPA - 8.31/10
Thesis title: *Vietnamese Punctuation Prediction Using Deep Neural Networks*.
Supervisor: Dr. Nguyen Thanh Binh.

Research Publications

Journal Articles

1 J. Park, A. G. Vien, M. Cha, **T. T. Pham**, H. Kim, and C. Lee, "Multiple transformation function estimation for image enhancement," *Journal of Visual Communication and Image Representation*, p. 103 863, 2023.

Conference Proceedings

1 **T. T. Pham**, T. T. N. Mai, and C. Lee, "Deep unfolded underwater image enhancement based on extreme channels prior," in *Proc. APSIPA Annual Summit and Conference (ASC)*, to be published, Oct. 2023.

2 **T. T. Pham**, T. T. N. Mai, and C. Lee, "Deep unfolding network with physics-based priors for underwater image enhancement," in *IEEE International Conference on Image Processing (ICIP)*, to be published, Oct. 2023.

3 **T. T. Pham**, T. T. N. Mai, and C. Lee, "Model-driven deep unfolding approach to underwater image enhancement," in *International Workshop on Advanced Imaging Technology (IWAIT) 2023*, SPIE, vol. 12592, 2023, p. 1 259 202.

4 **T. T. Pham**, T. T. N. Mai, and C. Lee, "Model-driven deep network-based rank minimization for image denoising," in *Summer Annual Conference of IEIE*, 2022, pp. 1328–1329.

5 **T. T. Pham**, T. T. N. Mai, and C. Lee, "Unrolling multi-channel weighted nuclear norm minimization for image denoising," in *2022 37th International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC)*, 2022, pp. 243–244. DOI: 10.1109/ITC-CSCC55581.2022.9894978.

6 N. T. Nguyen, **T. T. Pham**, T. X. Dang, *et al.*, "Malware detection using system logs," in *Proceedings of the 2020 on Intelligent Cross-Data Analysis and Retrieval Workshop*, 2020, pp. 9–14.

7 **T. Pham**, T. Dang, N. Nguyen, T. Ha, and B. T. Nguyen, "Early unsafety detection in autonomous vehicles," in *Computational Collective Intelligence: 12th International Conference, ICCCI 2020, Da Nang, Vietnam, November 30–December 3, 2020, Proceedings 12*, Springer, 2020, pp. 415–426.

8 **T. Pham**, T. Dang, N. Nguyen, V. T. Ha, and B. Nguyen, "Atgw: A machine learning framework for automation testing in game woody," in *Knowledge Innovation Through Intelligent Software Methodologies, Tools and Techniques: Proceedings of the 19th International Conference on New Trends in Intelligent Software Methodologies, Tools and Techniques (SoMeT_20)*, IOS Press, vol. 327, 2020, p. 413.

9 **T. Pham**, N. Nguyen, Q. Pham, H. Cao, and B. Nguyen, "Vietnamese punctuation prediction using deep neural networks," in *SOFSEM 2020: Theory and Practice of Computer Science: 46th International Conference on Current Trends in Theory and Practice of Informatics, SOFSEM 2020, Limassol, Cyprus, January 20–24, 2020, Proceedings 46*, Springer, 2020, pp. 388–400.

Skills

Languages	Vietnamese (native), English (good), Korea (beginner).
Technical skills	Python, SQL, Matlab, C/C++, Pytorch, Tensorflow, Keras, LaTeX, Microsoft Office.
Visualization	Tableau, Superset.

Awards and Achievements

2021 SRD Scholarship for International Students, Dongguk University, Seoul, Korea.

2019 Consolation Prize of the 21th annual Eureka Scientific Research Students Awards competition, Ho Chi Minh City, Vietnam.

References

Prof. Chul Lee
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Dongguk University (cilab.dongguk.edu),
10118 Engineering Building II, Dongguk University, 04620 Seoul, Korea.
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